

Vatsal Narula

908-864-1345 | vatsalnarula53@gmail.com | Portfolio | LinkedIn

EDUCATION

Rutgers University-New Brunswick

September 2021–May 2025

Bachelor of Science in Computer Science

- Relevant Courses: Systems Programming, Discrete Structures I & II, Computer Architecture, Data Structures, Algorithms, Deep Learning, Intro to Computational Robotics, Computer Security, Principles of Information and Data Management, Intro to Data Management.

EXPERIENCE

Machine Learning Research Assistant

May 2023 – Aug 2023

Rutgers University-New Brunswick

- Built an end-to-end **BERT4Rec** sequential recommender in **PyTorch**: temporal train/val/test split, sessionization, masked-item objective, sequence truncation/padding, and in-batch negative sampling.
- Implemented the data pipeline with vectorized loaders and a custom **Dataset/DataLoader** to emit token, position, and attention tensors with caching, materially improving preprocessing throughput and experiment turnaround.
- Ran controlled experiments with **AdamW**, cosine decay, early stopping, and ablations over max_seq_len, mask ratio, batch size, and learning rate.

Software Development Intern

May 2022 – Aug 2022

SISL

- Built and deployed a real-time conference room booking system using **Node.js/Express**, **MongoDB**, **EJS**, **AJAX** enabling 1,000+ employees across 4 departments to coordinate meetings with zero double-booking incidents post-launch.
- Exposed **REST** endpoints with validation and conflict checks; delivered the full-stack MVP in 6 weeks, improving booking throughput by 5× and reaching 90%+ adoption within 2 weeks.

PROJECTS

Distributed Peer-to-Peer Poker Platform | Go, TCP, gRPC, Protobuf, libp2p, Docker

- Built a decentralized peer-to-peer poker engine in **Go** with a deterministic state machine for deal, betting, and showdown over custom **TCP** and **gRPC/Protobuf**.
- Implemented **libp2p Kademlia DHT** for peer discovery and room matchmaking: advertised presence via Provide/FindProviders, used a bootstrap node, and enabled NAT traversal via relay and AutoNAT.
- Delivered **Docker**-based demos and deployment; designed a player registry and atomic counters to improve reliability and eliminate single points of failure.

AI Coding Assistant CLI | Go, Ollama, Docker

- Orchestrated OpenAI, Anthropic, Gemini, and Ollama through a Go CLI with interactive model selection, streaming, REPL memory, automatic codebase understanding, and a GitHub context loader.
- Built agentic automation to propose and execute shell commands with sandboxing via macOS seatbelt and Docker or Podman, with policy enforcement through allow and deny rules.
- Added regex routing to optimal models, JSONL tracing with a CLI dashboard, and diff-based apply with approvals and backups, Git integration, notifications, and local model management with Ollama.

Visual RAG System | FastAPI, React, Docker, Redis, AWS EKS

- Built a **REST API**-driven Visual RAG system using text and visual embeddings via ColPali with a **React + TypeScript** frontend for multi-document search over PDFs.
- Added async background processing with back-pressure and retries; leveraged **Redis** for caching and job status to keep latency predictable.
- Containerized the full stack with **Docker** and **NGINX**; deployed on **AWS EKS** with health checks and readiness probes for scalable orchestration.

ACHIEVEMENTS

Winner of SQC Quant Challenge, Rutgers University

- Developed a pairs trading strategy for cointegrated stocks using Kalman Filters and LSTM networks; achieved **Sharpe 1.5** and **Beta 0.02**.

LEADERSHIP

Mentor, UHAACS Hackathon, Rutgers — Guided 11 teams through ideation, design reviews, and debugging; coordinated judging criteria and ensured on-time submissions across tracks.

SKILLS

Programming Languages: Python, Java, Go, TypeScript, SQL

Frameworks & Libraries: Spring Boot, FastAPI, Flask, React, Django, Node.js, PyTorch, Pandas, Scikit-learn, EJS, AJAX, Material-UI, WebSocket

DevOps & Tools: Git, GitHub Actions, Docker, Kubernetes, Terraform, Prometheus, Grafana, NGINX

Cloud & Databases: AWS — EKS, EC2, S3, KMS; GCP — Pub/Sub; MongoDB, PostgreSQL, MySQL, Redis